User Management API

Relevant source files

- `Backend/STUDER/src/main/java/facu/studer/controllers/UserController.java`
- `Backend/STUDER/src/main/java/facu/studer/entities/User.java`
- `Backend/STUDER/src/main/java/facu/studer/exceptions/ApiError.java`
- `Backend/STUDER/src/main/java/facu/studer/exceptions/GlobalExceptionHandler.java`
- `Backend/STUDER/src/main/java/facu/studer/mappers/UserMapper.java`
- `Backend/STUDER/src/main/java/facu/studer/repositories/UserRepository.java`
- `Backend/STUDER/src/main/java/facu/studer/services/UserService.java`
- `Backend/STUDER/src/main/java/facu/studer/services/implementation/UserServiceImpl.java`

The User Management API provides the essential infrastructure for managing identity, profiles, and discovery within the STUDER platform. It handles the lifecycle of a user account—from registration and profile updates to soft deletion—and manages complex profile picture processing via integration with the media service.

API Endpoints

The `UserController` exposes endpoints under the `/api/v1/users` prefix. While registration is public, most operations require a valid JWT and enforce ownership checks, ensuring users can only modify their own data.

Method	Endpoint	Description	Auth Required
`POST`	`/register`	Registers a new user account.	No
`PUT`	`/`	Updates authenticated user profile (supports multipart/form-data).	Yes
`DELETE`	`/`	Soft deletes the authenticated user account.	Yes
`GET`	`/me`	Returns full profile data for the current user.	Yes

Method	Endpoint	Description	Auth Required
`GET`	`/username/{username}`	Returns public profile data for a specific user.	Yes
`GET`	`/search`	Paginated search for users by username.	Yes
`GET`	`/{id}`	Gets user data by ID (restricted to self).	Yes

Sources: `Backend/STUDER/src/main/java/facu/studer/controllers/UserController.java` `24-97`

Business Logic & Implementation

The `UserServiceImpl` class implements the core business logic, including data validation, password hashing, and interaction with external services.

Data Flow: User Update & Image Processing

When a user updates their profile with a new image, the backend acts as an orchestrator between the client and the Python-based Media Service.

1. **Request Reception**: `UserController` receives a `MultipartFile` and a `UserUpdateRequestDTO`
   `Backend/STUDER/src/main/java/facu/studer/controllers/UserController.java` `47-51`
2. **Media Service Call**: `UserServiceImpl` forwards the file to the Media Service at `${app.media.service.url}/upload-image/profile` using `RestTemplate`
   `Backend/STUDER/src/main/java/facu/studer/services/implementation/UserServiceImpl.java` `91-103`
3. **Variant Storage**: The Media Service returns a map of URLs for different image variants. The `User` entity stores these in four distinct fields
   `Backend/STUDER/src/main/java/facu/studer/entities/User.java` `70-81` :
   - `profilePictureOriginalUrl`: The raw uploaded file.
   - `profilePictureAvatarUrl`: 1080p high-quality version.
   - `profilePictureWebpUrl`: Optimized WebP format for performance.
   - `profilePictureThumbnailUrl`: 150x150 small crop for lists.
4. **Persistence**: The `UserRepository` saves the updated `User` entity
   `Backend/STUDER/src/main/java/facu/studer/services/implementation/UserServiceImpl.java` `120-121`

Soft Delete Pattern

Instead of removing rows from the database, STUDER uses a soft-delete mechanism. The `softDelete` method sets the `isActive` flag (inherited from `BaseEntity`) to `false`

`Backend/STUDER/src/main/java/facu/studer/services/implementation/UserServiceImpl.java` `132-134`

This preserves data integrity for related entities like discussions or messages while preventing the user from logging in.

Sources:

`Backend/STUDER/src/main/java/facu/studer/services/implementation/UserServiceImpl.java` `32-136`

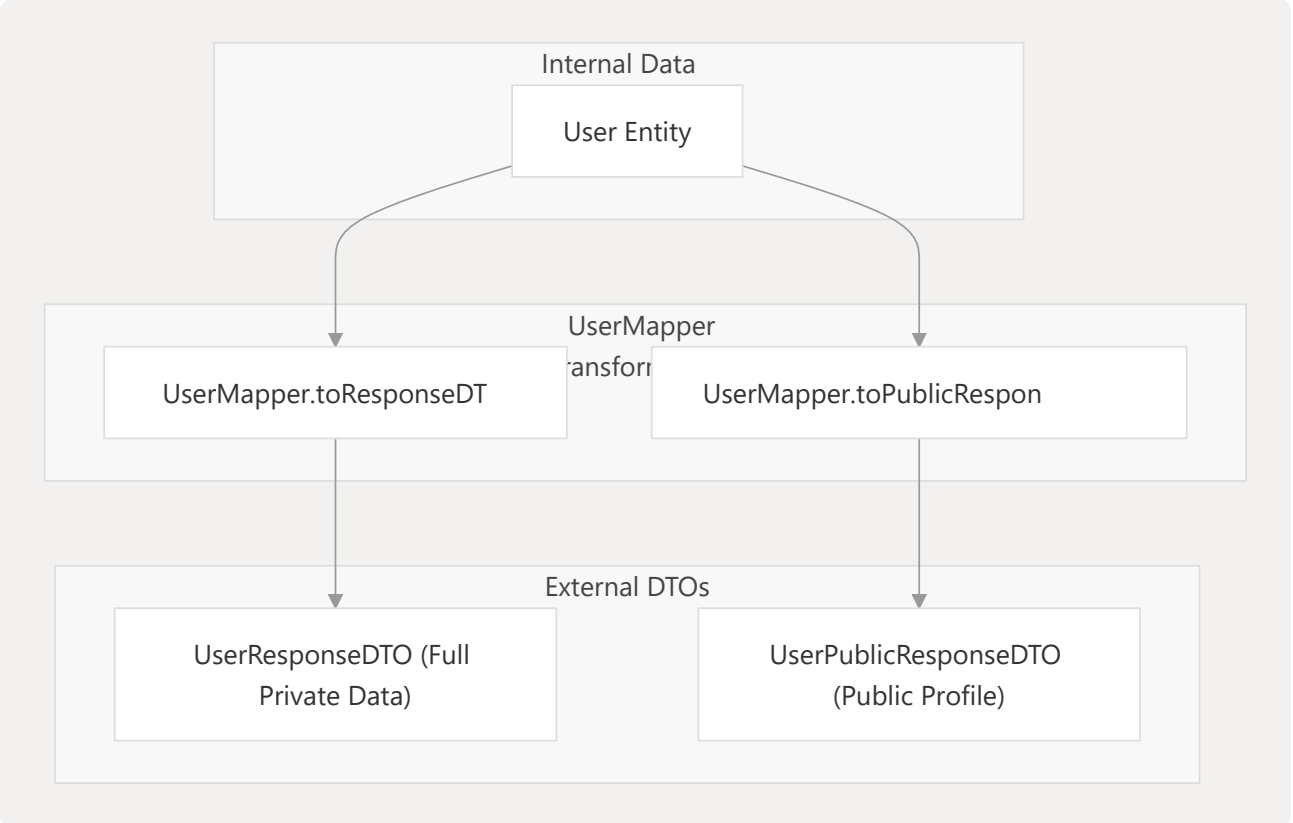
`Backend/STUDER/src/main/java/facu/studer/entities/User.java` `26-82`

Data Transformation (Mappers)

The `UserMapper` utility class handles the conversion between internal `User` entities and various Data Transfer Objects (DTOs) to ensure sensitive information (like hashed passwords) is never leaked to the frontend.

Mapping Architecture

Title: User Entity to DTO Mapping Flow



Sources: `Backend/STUDER/src/main/java/facu/studer/mappers/UserMapper.java 11-76`

Error Handling Contract

The API uses a centralized **`GlobalExceptionHandler`** to catch exceptions and return a standardized **`ApiError`** response. This contract ensures the frontend can consistently handle validation errors and internationalized messages.

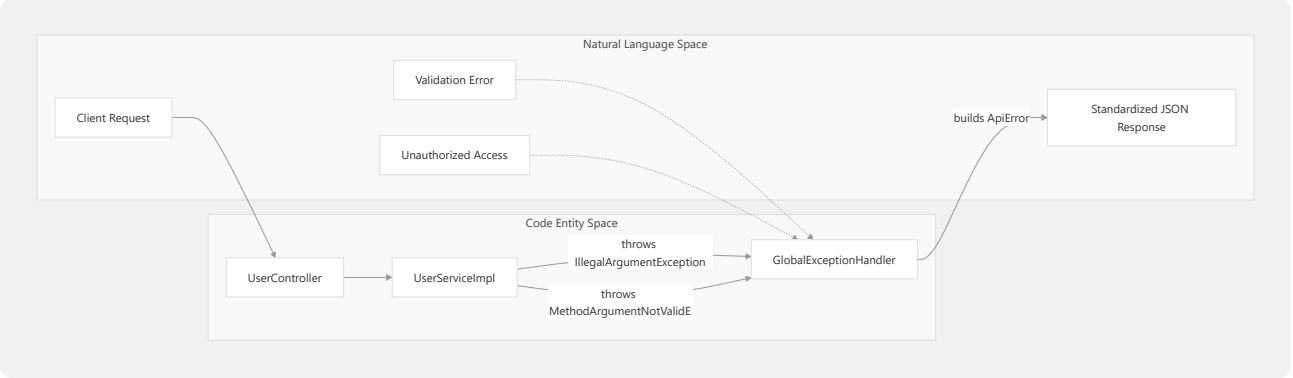
The ApiError Structure

The **`ApiError`** class `Backend/STUDER/src/main/java/facu/studer/exceptions/ApiError.java 17-47` includes:

- **`status`**: HTTP status code.
- **`code`**: A machine-readable string (e.g., **`user.email.unique`**) used as an i18n key.
- **`message`**: A human-readable, localized message resolved via **`MessageSource`**
  `Backend/STUDER/src/main/java/facu/studer/exceptions/GlobalExceptionHandler.java 93-104`
- **`fieldErrors`**: A map specifically for **`MethodArgumentNotValidException`**, linking specific fields (e.g., "email") to their validation failure reasons
  `Backend/STUDER/src/main/java/facu/studer/exceptions/GlobalExceptionHandler.java 31-53`

Exception Handling Logic

Title: Exception Resolution Process



Sources:

`Backend/STUDER/src/main/java/facu/studer/exceptions/GlobalExceptionHandler.java 21-107`

`Backend/STUDER/src/main/java/facu/studer/exceptions/ApiError.java 14-47`

Persistence Layer

The `UserRepository` interface extends `JpaRepository` and provides specific query methods for user lookup and uniqueness validation.

- **Uniqueness Checks**: `existsByEmail` and `existsByUsername` are used during registration and updates to prevent collisions
  `Backend/STUDER/src/main/java/facu/studer/repositories/UserRepository.java 20-27`
- **Search**: `findByUsernameContainingIgnoreCase` supports the user discovery feature with paginated results
  `Backend/STUDER/src/main/java/facu/studer/repositories/UserRepository.java 49-50`

Sources: `Backend/STUDER/src/main/java/facu/studer/repositories/UserRepository.java 14-50`